Ayush Sinha

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Education

National Institute of Technology, Rourkela **September 2023 – Present**
Bachelor of Technology in Civil Engineering (CGPA: 7.84/10) *Rourkela, Odisha*

Relevant Coursework

- Data Structures
- Algorithms
- Operating Systems
- Object Oriented Programming
- Database Management Systems
- Computer Networks

Work Experience

Quickserve **May 2026 – July 2026**
SDE Intern | React, Node.js, Express.js *Certificate*

- Developed an internal service-tracking dashboard in **React**, giving the operations team a single view to monitor, filter, and manage customer requests efficiently.
- Developed **REST API** endpoints in **Node.js** and **Express.js** to serve data for the dashboard's different components, connecting the **React** frontend to the backend.

Projects

Distributed API Rate Limiter **August 2026**
Java, Spring Boot, Redis, Kafka, Docker, PostgreSQL *GitHub*

- Engineered a **containerized rate-limiting microservice** in **Spring Boot**, orchestrating a **6-node** Docker environment spanning PostgreSQL, Redis, and Apache Kafka.
- Implemented high-throughput API traffic control with the **Token Bucket algorithm** backed by atomic **Redis Lua scripts**, achieving **<1ms** processing times.
- Instrumented the application with **Prometheus** and created **Grafana** dashboards to visualize real-time API throughput, HTTP 429 rejection rates, and system latency at a **15-second** scrape interval.

MailSense AI **June 2026**
Java, Spring Boot, React, Chrome Extension, Gemini API, RAG *GitHub*

- Built a **Chrome extension** (**Manifest V3**, **JavaScript**) that adds a Generate Reply button to Gmail's compose window, detecting compose boxes with a **MutationObserver** and inserting drafts in **7** tone options.
- Implemented **retrieval-augmented generation (RAG)** in **Spring Boot** using **768-dimension** Gemini embeddings and **cosine similarity**, retrieving the top **5** past emails so drafts match the user's writing style.
- Built **4 REST API** endpoints in **Java 21** that call the **Google Gemini API**, with retries on HTTP 429/500/503 errors and an automatic fallback model to keep replies working during API overload.

Sleepyhead Studios Website **November 2025**
Next.js 14, React 19, Framer Motion, Lenis, Vercel *GitHub | Website*

- Architected a production-ready web platform with **Next.js 14** and **React 19**, integrating **Framer Motion** to deliver responsive UI components across **5** core routes.
- Improved frontend performance by leveraging **Next.js Server-Side Rendering (SSR)**, achieving perfect **100** Lighthouse Performance and SEO scores to maximize discoverability.
- Streamlined production releases with a Git-based workflow integrated directly into **Vercel**, managing custom domain routing and **DNS configuration** for the live application.

Achievements / Certifications

- **Open Source Contributor – Lichess** *GitHub* | **June 2026**
  Authored **3 pull requests** merged into production, improving opening classifications and transposition data.
- **HACKINNO Hackathon** *Project* | **September 2024**
  Secured **2nd place** among **50+ teams**, prototyping the landing page and core logic for an eco-builder game.
- **500+ DSA Problems Solved** *LeetCode* | **Ongoing**
  Active across **LeetCode**, **Codeforces**, and other platforms, maintaining a **1700** LeetCode contest rating.

Technical Skills

- **Languages**: Java, JavaScript, TypeScript, Python, SQL
- **Developer Tools**: VS Code, IntelliJ IDEA, Git, Docker, Prometheus, Grafana, Postman
- **Frameworks/Libraries**: Spring Boot, React, Next.js, PostgreSQL, MongoDB, Redis, Apache Kafka, REST APIs, Microservices

Extracurricular Activities

President, The Knights Club NITR **May 2025 – April 2026**

- Led executive board operations as President, mentoring members to grow club participation to **50+ students** and securing **gold at Vriddhi'23** and **bronze at Parakram'24**.